

Credit Risk Project Report And Analysis

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Executive Summary

Throughout this project, I developed a credit risk scoring model in order to predict loan defaulting using an online Kaggle data set which contains 33,851 loan records. The goal of this project was to determine which borrower characteristics/features were the most important or had the most weight when predicting default risk. Another essential part of this project was producing a highly trustworthy machine learning model that can flag high risk applicants.

Key Findings:

- The overall default rate in the entire data set was 22%
- Across the several models, DTI (debt-to-income ratio) is the strongest and most important predictor of default.
- Borrowers with prior defaults in the system are significantly more likely to default again
- Borrowers who rent are more likely to default compared to borrowers who own, have a mortgage, or have any other type of home ownership
- Lower income and shorter employment history correlate with higher default risk

Model Performance

The three machine learning models of Logistic Regression, Random Forest, and XGBoost were evaluated. Through metrics such as accuracy, precision, recall, and AUC-ROC, we compared all three models with XGBoost coming out on top with the best overall performance. However, the logistic regression model did show interesting insights regarding each borrower's characteristic's impact on default probability. Exact results will be shown later.

Recommendations:

- Flag applicants with DTI above 40% and any prior default history for additional review
- Consider income-to-loan thresholds as a first-pass filter
- Use the XGBoost model for automated risk scoring in the loan approval pipeline

Limitations:

One key limitation found was the class imbalance found in the Kaggle dataset. When looking at the dataset, I found that approximately 78% of the borrowers did not default while 22% did default. When training, models will see way more "good" loans than "bad" loans and a lazy model can just guess that all borrowers are not going to default and it will automatically get a 78% accuracy while catching 0 actual defaults. We can use SMOTE on training data to balance the dataset.

Introduction/Background

The Business Problem

Loan defaults represent one of the most significant financial risks facing lending institutions. When a borrower fails to repay a loan, lenders lose not only the principal amount but also the expected interest income, while incurring additional costs for collections and write-offs. For banks and financial institutions, accurately predicting which borrowers are likely to default is important for preserving profitability and portfolio risk.

The real challenge is finding the right balance. Playing it too safe/conservative leads to rejecting creditworthy applicants and losing potential revenue. Being too lenient leads to approving higher risk borrowers who are likely to default. An effective credit risk model helps lenders find this balance by quantifying the probability of default based on borrower and loan characteristics.

Project Objective

This project aims to answer a fundamental question in credit risk analysis: What borrower and loan characteristics most strongly predict default, and how can lenders use this information to make better risk-adjusted lending decisions?

To answer this question, throughout this project I combined SQL-based exploratory analysis and machine learning modeling - combining all of this with data visualization at the end.

- SQL Analysis - Segment borrowers by various characteristics (income, age, debt-to-income ratio, home ownership, etc.) and calculate default rates for each segment to identify patterns and risk factors.
- Machine Learning - Train and compare multiple classification models (Logistic Regression, Random Forest, XGBoost) to predict default probability and identify the most important predictive features.
- Visualization - Build interactive Tableau dashboards and generate plots demonstrating the model metrics that communicate findings to business stakeholders and enable data-driven lending decisions.

Dataset Overview

The analysis uses a credit risk dataset from Kaggle containing 32,581 loan records with 12 features capturing borrower demographics, financial characteristics, and loan details.

Dataset Features

Feature	Description	Type
person_age	Borrower age in years	Numeric
person_income	Annual income in dollars	Numeric
person_home_ownership	Housing status: RENT, OWN, MORTGAGE, OTHER	Categorical
person_emp_length	Employment length in years	Numeric
loan_intent	Purpose: EDUCATION, MEDICAL, DEBTCONSOLIDATION, HOMEIMPROVEMENT, PERSONAL, VENTURE	Categorical
loan_grade	Risk grade A-G (low to high risk)	Categorical
loan_amnt	Loan amount requested	Numeric
loan_int_rate	Interest rate percentage	Numeric
loan_status	Target: 0 = paid, 1 = defaulted	Binary
loan_percent_income	DTI ratio (loan/income)	Numeric
cb_person_default_on_file	Prior default: Y/N	Categorical
cb_person_cred_hist_length	Credit history length in years	Numeric

Class Imbalance

The dataset contains 78% non-defaults and 22% defaults. This class imbalance affects model training because models tend to favor the majority class to maximize accuracy. As said before a stupid model predicting "no default" for everyone achieves 78% accuracy while catching zero actual defaults. Cross-validation scores show higher variance for minority class metrics (recall)

SQL Analysis Approach

Basic Analysis (analysis.sql):

Created views calculating default rates segmented by:

Age brackets (18-24, 25-32, 33-45, 46-55, 55+)

Income brackets (\$0-25k, \$25k-50k, \$50k-75k, \$75k-100k, \$100k-150k, \$150k+)

Home ownership (rent, own, mortgage, other)

Employment length (0-2 yrs, 3-5 yrs, 6-10 yrs, 10+ yrs)
Loan intent (education, medical, debt consolidation, etc.)
DTI brackets (0-10%, 10-20%, 20-30%, 30-40%, 40-50%, 50%+)
Prior default history (Y/N)
Credit history length

Advanced Analysis (analysis_2.sql):

Demonstrated intermediate-advanced SQL skills:
CTEs (Common Table Expressions): Multi-step queries for risk scoring
Window Functions: NTILE() to create risk deciles, SUM() OVER() for cumulative calculations
Risk Scoring Model: Combined 6 risk factors into a single risk flag count

Machine Learning Approach

Data Preprocessing:

Dropped null values (primarily in employment length)
One-hot encoded categorical variables
Split data 80/20 into training and test sets

Models Trained:

Logistic Regression — Baseline model, interpretable coefficients
Random Forest — Ensemble of decision trees, handles non-linear relationships
XGBoost — Gradient boosting, typically best performance

Evaluation Metrics:

Accuracy: Overall correct predictions (limited usefulness due to class imbalance)
Precision: When predicting default, how often correct
Recall: Of actual defaults, how many caught (critical for credit risk)
AUC-ROC: Overall ability to distinguish defaulters from non-defaulters

Cross-Validation:

5-fold cross-validation was used to assess model stability and generalization.